ME 5659 Project Proposal

Control of a Two-Dimensional Submarine System

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For our project, we will consider the dynamics and controls of a two-dimensional submarine system. We will limit the scope of the plant dynamics to be a function of latitude (x), depth (y), and pitch angle (θ), with corresponding input signals rudder force and ballast force (see diagram below). After further research, we may also consider higher order terms like hydrostatic pressure and drag and explore the modes of control typically used in the field (torque control vs velocity control, etc.). We plan to design two to three different controllers that target different goal objectives; some examples may include time, fuel, or thrust optimization.

We will define a set of system parameters and global assumptions that our model operates under, of which our control designs will be subject to, and evaluate their performances in terms of stability, transient behavior, settling time, and other imposed metrics specific to a submarine system, e.g. maximum linear acceleration for safety considerations. We plan to use MATLAB Simulink as the primary simulation and analysis platform. After evaluation, we will conclude with discussion of the strengths and weaknesses of each approach and how our design choices address the control problem that they aim to solve, as well as the challenges we faced during the process.

Visualization of an example scenario for our proposed submarine system.

